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Mentoring at scale: Acknowledging Implicit Mentoring

Anita Sarma, Oregon State University

@asarma

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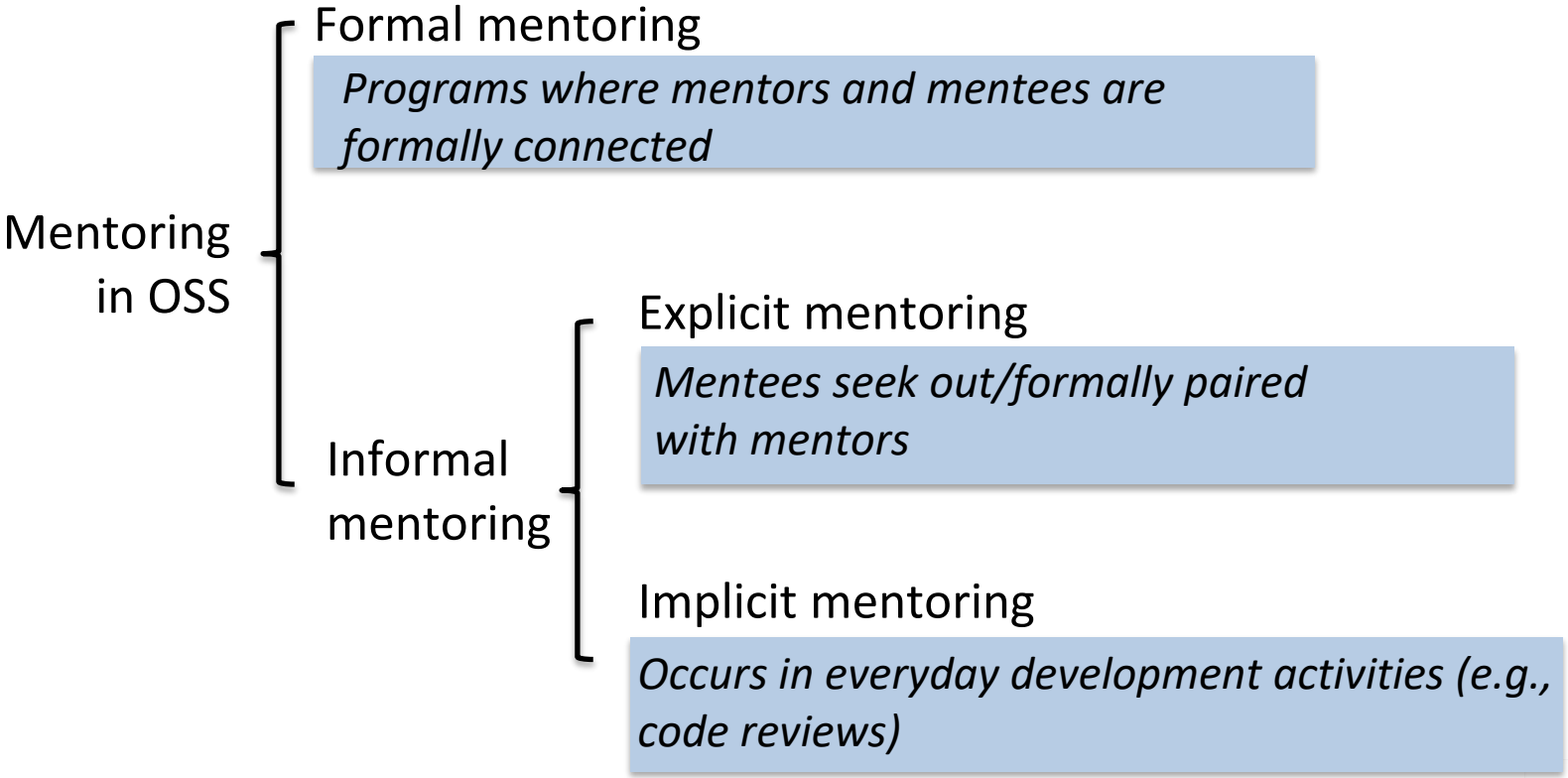
Mentoring in OSS



Implicit Mentoring

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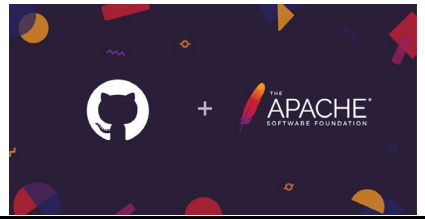
Implicit mentoring

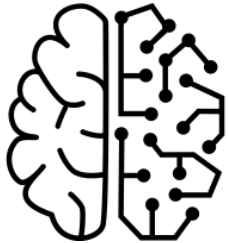
Implicit mentoring can be defined as “mentoring that occurs in *everyday development activities* such as code reviews, where a mentor provides an *underlying explanation* when providing suggestions, instructions, or mechanisms to address errors”

We define, we mine, we classify

37 repositories



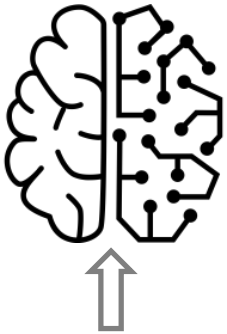
97,444
Pull Request (PR)
comments
11,634 contributors



Machine learning
classifier

Training data through
manual labeling

Training the ML classifier



Explanations accompanying GitHub PR Comments

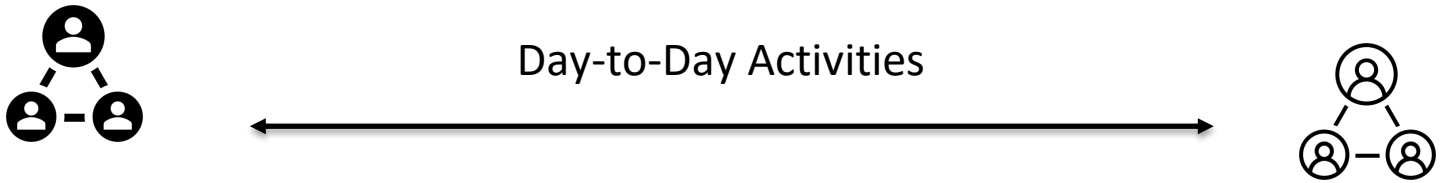
Rule 1

Rule 2

Rule 3

Mentoring Action	PR_comment sample
Instruction	M-284: "...run [tool] on the project before creating a PR. You would have noticed [problem]..."
Suggestion	B-553: " I would still duplicate [action] like I did in [certain PR] because it's widely used in [tests]. Maybe this could be removed after [situation]."
Mechanisms to fix errors	I-1376: " Would you mind just doing [action] again to kick off [framework]? I think [framework] is just not happy when it has a lot of loads."

Implicit mentoring is widespread



- 30.27% of Pull-Request included implicit mentoring.
- 25.24% of contributors served as implicit mentors.

Features of implicit mentoring

Of the 29,502 PRs: dyadic (65.15%), triads (22.79%) >quadrads (12.06%). “it takes a village...”

It is not just top to down, also including bottom up (13.08%), and peer to peer (<6 months diff, 34.14%).

Implicit mentoring is interest driven; can it be drive-by mentoring?

More than dyadic

More than top to down

Is topical

Does gender play a role?

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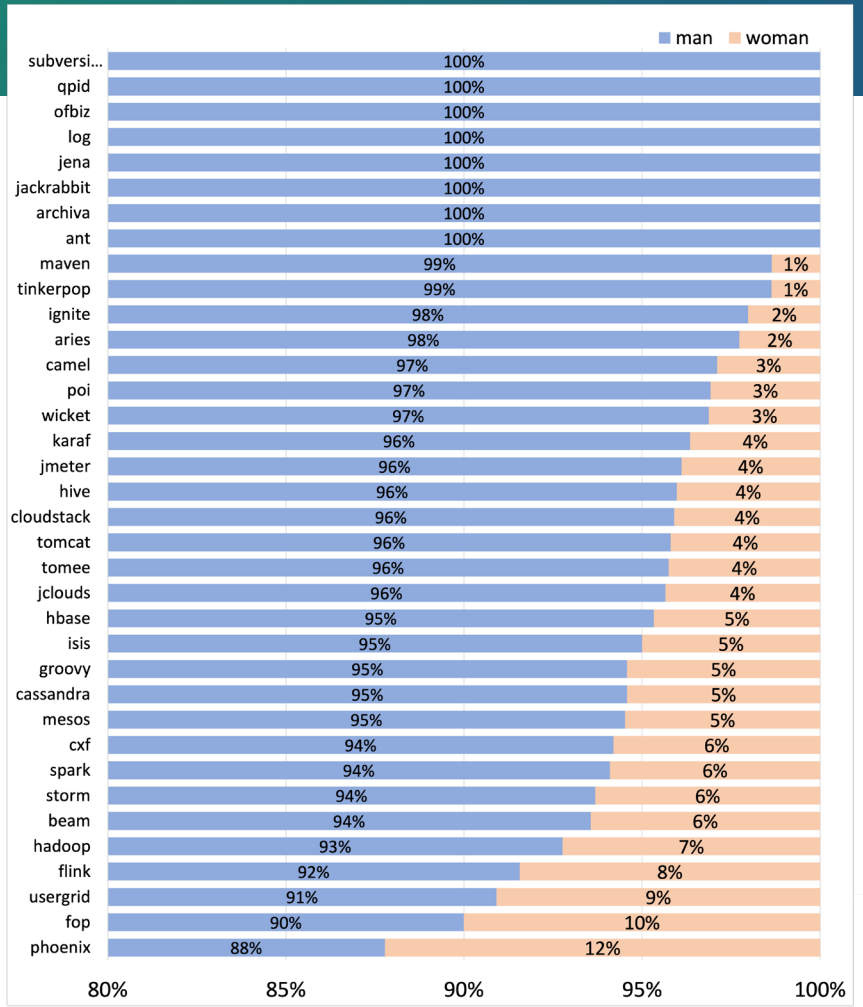
Identifying gender

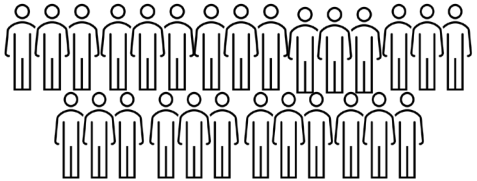
Used NameSorML API

- First & last names
- Geographical location
- >90% confidence



https://www.namsor.com/?_ga=2.167230650.527668813.1632441804-97789108.1632441804

Gender and implicit mentoring?



Men: 94.43%

Women: 5.57%

Implicit mentors: 94.70%

Implicit mentors: 5.30%

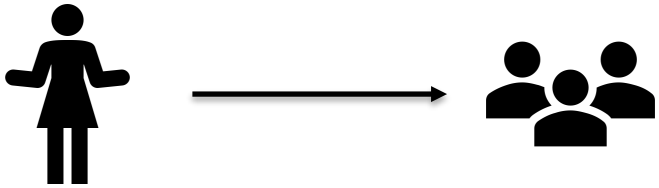
- Men perform implicit mentoring more often.
- Proportionality test (7% more, p-value<0.001), but low effect size

Homophily preferences of implicit mentoring

	Overall	Top -> Down		Peer -> Peer		Bottom -> Up	
W -> W	316	55	0.20%	209	1.14%	52	0.89%
W -> M	453	125	0.46%	179	0.98%	149	2.55%
M -> W	1642	839	3.08%	735	4.02%	68	1.17%
M -> M	48982	26256	96.26%	17162	93.86%	5564	95.39%
Total	51393	27275		18285		5833	

Homophilic mentoring surpassed cross-gender mentoring for top->down, peer -> peer, and bottom -> up mentoring with large effect size.

Do mentors “reach across the aisle”?



In the few cases of cross-gender implicit mentoring women tend to cross gender boundaries more often (56%) than men ($p\text{-value}<0.001$, $d=1.39$) .

Discussion

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Discussion

- Why is there homophily in implicit mentoring via code review?
  - Does this mean fewer women get mentored?
- How to acknowledge implicit mentoring?
  - Would we retain more mentors if they are acknowledged formally?
- What other “invisible work” are we missing?
  - Would acknowledging invisible /non-coding work attract and retain more women in OSS?

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Thank you!

anita.sarma@oregonstate.edu
@asarma

Thank you to Interviewees and team and

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